Programming in C

FILES

In C programming, file handling is a crucial aspect that allows programs to store and retrieve data from files on a computer's storage device. Without file handling, any data generated during program execution would be lost once the program terminates. File handling ensures that data
persists beyond the program's lifecycle, enabling applications to maintain state, store user inputs, and manage large datasets efficiently.

C supports two primary types of files:

1**. Text Files:** These files store data in a human-readable format. Each line of text is terminated by a newline character.
Text files are commonly used for storing configuration settings, logs, or simple data records.

2. **Binary Files:** Unlike text files, binary files store data in a format that is directly readable by the computer.
They can hold complex data structures and are more efficient for storing large volumes of data, such as images, audio files, or serialized objects.

File Operations:

- Opening a File: Before performing any operations on a file, it must be opened using the fopen() function.
  This function associates a file with a file pointer.

- Reading from a File: Once a file is opened, data can be read using functions like fgetc(), fgets(), or fread(),
  depending on the type and structure of the data.

- Writing to a File: Data can be written to a file using functions such as fputc(), fputs(), or fwrite().
  These functions allow for writing characters, strings, or binary data,

respectively.

- Closing a File: After all operations on a file are complete, it should be closed using the fclose() function.
This ensures that all data is properly written and resources are released.

File Modes:

- "r": Open for reading. The file must exist.
- "w": Open for writing. Creates a new file or truncates an existing file.
- "a": Open for appending. Data is written at the end of the file.
- "r+": Open for reading and writing. The file must exist.
- "w+": Open for reading and writing. Creates a new file or truncates an existing file.
- "a+": Open for reading and appending. Data is written at the end.

CLASSIFICATION OF DATA STRUCTURE

Based on how the data items are operated, it will classify into two broad categories.

- Primitive Data Structure
- Non-Primitive Data Structure

Primitive Data Structure: These are basic DS and the data items are operated closest to the machine level instruction. Example: integer, characters, strings, pointers and double.

Non-Primitive Data Structure: These are more sophisticated DS and are derived from primitive DS. Here data items are not operated closest to machine level instruction. It emphasize on structuring of a group of homogeneous (same type) or heterogeneous (different type) data items.

Linear Data Structure: In which the data items are stored in sequence order. Example: Arrays, Linked Lists, Stacks and Queues Non Linear Data Structure: In Non-Linear data structures, the data items are not in sequence. Example: Trees, Graphs.

File handling in C is essential for applications that require persistent data storage, including databases, text editors, and data logging systems. By understanding file types, operations, and modes, programmers can create efficient and reliable applications that interact seamlessly with external data sources.

Further elaboration and examples can be included with real-world applications such as storing student records, managing inventory, or logging system events. The versatility of C file handling makes it an indispensable concept for all aspiring programmers.

MCQ QUESTIONS

1. Which function is used to open a file in C?

a) fopen()

b) fclose()

c) fread()

d) fputc()

2. What does the mode "r" signify when opening a file?

a) Open for writing

b) Open for reading

c) Open for appending

d) Open for reading and writing

3. Which of the following functions reads a single character from a file?

a) fputs()

b) fgetc()

c) fgets()

d) fprintf()

4. What is the difference between text and binary files?

a) Text files store human-readable characters; binary files store data in machine format

b) Binary files are always smaller than text files

c) Text files cannot be opened in C

d) Binary files are read using fputs()

5. Which function closes a file after operations are complete?

a) fclose()

b) fseek()

c) fread()

d) fopen()

6. What happens if you open a file in "w" mode and the file already exists?

a) File is read-only

b) File is appended

c) File is overwritten

d) Error occurs

7. Which function is used to read a line from a text file?

a) fgetc()

b) fgets()

c) fwrite()

d) fprintf()

8. In which mode can you both read and write to an existing file?
a) "r"
b) "w"
c) "r+"
d) "a"

9. What is the purpose of a file pointer?
a) It points to a memory location of a variable
b) It stores the contents of the file
c) It acts as a reference to a file for performing operations
d) It deletes the file after reading

10. Which function writes a single character to a file?
a) fputc()
b) fputs()
c) fprintf()
d) fread()

11. Which function writes a string to a file?
a) fputs()
b) fgetc()
c) fread()
d) fclose()

12. Which mode writes at the end of the file without erasing existing content?
a) "w"
b) "a"
c) "r"
d) "w+"

13. Which of the following is NOT a file operation in C?
a) Reading
b) Writing
c) Sorting
d) Closing

14. What type of file stores data in a format directly readable by the computer?
a) Text file
b) Binary file
c) CSV file
d) XML file

15. Which function reads a block of data from a binary file?
a) fputs()

b) fread()

c) fgetc()

d) fprintf()

16. What happens if fclose() is not called after file operations?

a) Nothing happens

b) Data may not be saved and resources may be leaked

c) File is automatically deleted

d) File is converted to binary

17. Which of the following is a valid mode for reading and appending?

a) "r"

b) "w+"

c) "a+"

d) "r+"

18. Which function writes formatted data to a file?

a) fscanf()

b) fprintf()

c) fputs()

d) fgetc()

19. A file in C is

a) Temporary memory storage

b) A collection of data stored on disk with a name

c) Only used for images

d) Only used for text

20. Which of the following is TRUE about text files?

a) They can store complex memory structures

b) They are human-readable

c) They cannot be opened in C

d) They are always smaller than binary files